

Weekly Topical Worksheet Pack**CANDIDATE****Urwah****SUBJECT****Chemistry -- Cambridge International A Level (9701)****TOTAL MARKS (WEEK)****168 marks****HOW TO USE THIS PACK**

One short set of questions per session, matching that day's topic from the schedule.

Answer in the spaces / on the lines provided. Marks for each part are shown in [brackets].

Mark schemes for the whole week are collected at the end of the pack.

These are original questions written to match Cambridge's exact style and command words — not reproductions of real past paper text.

THIS WEEK AT A GLANCE

TOPIC 10 — TOPIC 10.1: GROUP 2 (36 marks)

TOPIC 11 — TOPIC 11.1: GROUP 17 (50 marks)

TOPIC 12 — TOPIC 12.1: NITROGEN AND SULFUR (32 marks)

TOPIC 13 — TOPIC 13.1: AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY (50 marks)

MENTORA ACADEMY · EXCELLENCE IN EDUCATION.**TOPICAL PRACTICE — WEEK 2****Inorganic Chemistry -- Topics 9–12: Periodicity to Nitrogen & Sulfur**

AS Level Chemistry 9701 -- 2025–2027 Syllabus

Question Pack

TOPIC 10 — TOPIC 10.1: GROUP 2

- 1** **GROUP 2 — 9701/21/O/N/22/Q8(c)** Syllabus 10.1
...
(a) State the flame colour of barium. [1]
- 2** **GROUP 2 — 9701/24/F/M/22/Q6(b)** Syllabus 10.1
...
(a) Explain the trend in thermal stability of Group 2 carbonates. [3]
- 3** **GROUP 2 — 9701/22/F/M/22/Q7(a)** Syllabus 10.1
...
(a) Write an equation for the thermal decomposition of magnesium nitrate. [2]
- 4** **GROUP 2 — 9701/21/M/J/19/Q7(b)** Syllabus 10.1
...
(a) Compare the reactivity of magnesium and barium with water. [2]
- 5** **GROUP 2 — 9701/21/F/M/25/Q1(c)** Syllabus 10.1
...
"[DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates]"
(a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]
- 6** **GROUP 2 — 9701/23/F/M/24/Q4(b)** Syllabus 10.1
...
(a) State the flame colour of barium. [1]
- 7** **GROUP 2 — 9701/22/M/J/25/Q8(b)** Syllabus 10.1
...
(a) Explain the trend in thermal stability of Group 2 carbonates. [3]
- 8** **GROUP 2 — 9701/24/F/M/25/Q4(a)** Syllabus 10.1
...
(a) Write an equation for the thermal decomposition of magnesium nitrate. [2]
- 9** **GROUP 2 — 9701/22/M/J/21/Q8(c)** Syllabus 10.1
...
(a) Compare the reactivity of magnesium and barium with water. [2]
- 10** **GROUP 2 — 9701/21/O/N/19/Q4(b)** Syllabus 10.1
...
"[DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates]"
(a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]

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- 11** **GROUP 2 – 9701/24/O/N/20/Q2(c)** Syllabus 10.1
....
(a) State the flame colour of barium. [1]
- 12** **GROUP 2 – 9701/22/F/M/20/Q8(c)** Syllabus 10.1
....
(a) Explain the trend in thermal stability of Group 2 carbonates. [3]
- 13** **GROUP 2 – 9701/24/F/M/25/Q3(b)** Syllabus 10.1
....
(a) Write an equation for the thermal decomposition of magnesium nitrate. [2]
- 14** **GROUP 2 – 9701/22/O/N/24/Q5(c)** Syllabus 10.1
....
(a) Compare the reactivity of magnesium and barium with water. [2]
- 15** **GROUP 2 – 9701/22/M/J/19/Q6(c)** Syllabus 10.1
" [DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates] "
(a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]
- 16** **GROUP 2 – 9701/21/M/J/23/Q8(a)** Syllabus 10.1
....
(a) State the flame colour of barium. [1]
- 17** **GROUP 2 – 9701/23/F/M/21/Q7(a)** Syllabus 10.1
....
(a) Explain the trend in thermal stability of Group 2 carbonates. [3]
- 18** **GROUP 2 – 9701/22/O/N/20/Q2(b)** Syllabus 10.1
....
(a) Write an equation for the thermal decomposition of magnesium nitrate. [2]
- 19** **GROUP 2 – 9701/24/M/J/23/Q9(c)** Syllabus 10.1
....
(a) Compare the reactivity of magnesium and barium with water. [2]
- 20** **GROUP 2 – 9701/23/O/N/20/Q10(c)** Syllabus 10.1
" [DIAGRAM: Experimental setup showing thermal decomposition of Group 2 carbonates] "
(a) Describe the trend in solubility of Group 2 hydroxides down the group. [1]

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TOPIC 11 — TOPIC 11.1: GROUP 17

21 GROUP 17 — 9701/21/F/M/19/Q3(b) Syllabus 11.2

""

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

22 GROUP 17 — 9701/24/M/J/25/Q9(a) Syllabus 11.1

""

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

23 GROUP 17 — 9701/22/F/M/20/Q2(c) Syllabus 11.2

""

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

24 GROUP 17 — 9701/22/O/N/21/Q10(a) Syllabus 11.2

""

- (a) Explain the use of chlorine in water treatment. [2]

25 GROUP 17 — 9701/21/O/N/21/Q5(a) Syllabus 11.2

""

- (a) Describe the trend in boiling points of the halogens. [2]

26 GROUP 17 — 9701/24/F/M/24/Q10(c) Syllabus 11.1

"Fig. 2.1 shows the trend in boiling points of the halogens."

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

27 GROUP 17 — 9701/21/F/M/22/Q5(a) Syllabus 11.1

""

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

28 GROUP 17 — 9701/22/O/N/20/Q4(b) Syllabus 11.3

""

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

29 GROUP 17 — 9701/21/M/J/19/Q2(c) Syllabus 11.3

""

- (a) Explain the use of chlorine in water treatment. [2]

30 GROUP 17 — 9701/21/O/N/25/Q5(b) Syllabus 11.3

""

- (a) Describe the trend in boiling points of the halogens. [2]

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- 31** **GROUP 17 – 9701/24/F/M/21/Q5(b)** Syllabus 11.2
-
- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]
- 32** **GROUP 17 – 9701/22/M/J/24/Q9(a)** Syllabus 11.1
- "Fig. 2.1 shows the trend in boiling points of the halogens."
- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]
- 33** **GROUP 17 – 9701/21/M/J/19/Q4(c)** Syllabus 11.2
-
- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]
- 34** **GROUP 17 – 9701/24/F/M/20/Q1(c)** Syllabus 11.3
-
- (a) Explain the use of chlorine in water treatment. [2]
- 35** **GROUP 17 – 9701/24/O/N/23/Q8(b)** Syllabus 11.3
-
- (a) Describe the trend in boiling points of the halogens. [2]
- 36** **GROUP 17 – 9701/24/M/J/20/Q8(b)** Syllabus 11.2
-
- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]
- 37** **GROUP 17 – 9701/23/F/M/21/Q4(a)** Syllabus 11.1
-
- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]
- 38** **GROUP 17 – 9701/22/F/M/25/Q8(a)** Syllabus 11.3
- "Fig. 2.1 shows the trend in boiling points of the halogens."
- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]
- 39** **GROUP 17 – 9701/23/M/J/19/Q4(a)** Syllabus 11.3
-
- (a) Explain the use of chlorine in water treatment. [2]
- 40** **GROUP 17 – 9701/21/M/J/21/Q9(c)** Syllabus 11.2
-
- (a) Describe the trend in boiling points of the halogens. [2]

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41 **GROUP 17 – 9701/22/F/M/21/Q6(b)** Syllabus 11.1

""

- (a) Explain why silver chloride dissolves in dilute aqueous ammonia but silver iodide does not. [2]

42 **GROUP 17 – 9701/21/O/N/20/Q5(a)** Syllabus 11.1

""

- (a) Write an equation for the reaction of chlorine with cold aqueous sodium hydroxide. [2]

43 **GROUP 17 – 9701/22/M/J/23/Q6(b)** Syllabus 11.2

""

- (a) Suggest why adding aqueous silver nitrate followed by aqueous ammonia is a good test for halide ions. [2]

44 **GROUP 17 – 9701/22/M/J/23/Q10(a)** Syllabus 11.4

"Fig. 2.1 shows the trend in boiling points of the halogens."

- (a) Explain the use of chlorine in water treatment. [2]

45 **GROUP 17 – 9701/21/F/M/22/Q4(c)** Syllabus 11.4

""

- (a) Describe the trend in boiling points of the halogens. [2]

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TOPIC 12 — TOPIC 12.1: NITROGEN AND SULFUR

46 NITROGEN AND SULFUR — 9701/23/M/J/22/Q6(c) Syllabus 12.1

...

- (a) Describe the environmental consequences of sulfur dioxide emissions. [2]

47 NITROGEN AND SULFUR — 9701/21/F/M/25/Q1(a) Syllabus 12.1

...

- (a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]

48 NITROGEN AND SULFUR — 9701/24/F/M/21/Q9(c) Syllabus 12.1

...

- (a) Explain how a catalytic converter removes NO from exhaust gases. [2]

49 NITROGEN AND SULFUR — 9701/22/F/M/20/Q8(b) Syllabus 12.1

...

- (a) State a natural source of sulfur dioxide in the atmosphere. [1]

50 NITROGEN AND SULFUR — 9701/24/M/J/24/Q6(b) Syllabus 12.1

"[DIAGRAM: Formation of acid rain from atmospheric pollutants]"

- (a) Explain how nitrogen oxides are formed in a car engine. [2]

51 NITROGEN AND SULFUR — 9701/21/O/N/22/Q6(b) Syllabus 12.1

...

- (a) Describe the environmental consequences of sulfur dioxide emissions. [2]

52 NITROGEN AND SULFUR — 9701/22/F/M/25/Q3(b) Syllabus 12.1

...

- (a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]

53 NITROGEN AND SULFUR — 9701/21/F/M/21/Q1(b) Syllabus 12.1

...

- (a) Explain how a catalytic converter removes NO from exhaust gases. [2]

54 NITROGEN AND SULFUR — 9701/24/F/M/23/Q3(c) Syllabus 12.1

...

- (a) State a natural source of sulfur dioxide in the atmosphere. [1]

55 NITROGEN AND SULFUR — 9701/23/F/M/23/Q2(b) Syllabus 12.1

"[DIAGRAM: Formation of acid rain from atmospheric pollutants]"

- (a) Explain how nitrogen oxides are formed in a car engine. [2]

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- 56 NITROGEN AND SULFUR – 9701/22/M/J/20/Q2(a)** Syllabus 12.1
....
(a) Describe the environmental consequences of sulfur dioxide emissions. [2]
- 57 NITROGEN AND SULFUR – 9701/21/F/M/24/Q5(c)** Syllabus 12.1
....
(a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]
- 58 NITROGEN AND SULFUR – 9701/23/F/M/21/Q10(b)** Syllabus 12.1
....
(a) Explain how a catalytic converter removes NO from exhaust gases. [2]
- 59 NITROGEN AND SULFUR – 9701/24/F/M/25/Q9(b)** Syllabus 12.1
....
(a) State a natural source of sulfur dioxide in the atmosphere. [1]
- 60 NITROGEN AND SULFUR – 9701/23/F/M/23/Q7(b)** Syllabus 12.1
" [DIAGRAM: Formation of acid rain from atmospheric pollutants] "
(a) Explain how nitrogen oxides are formed in a car engine. [2]
- 61 NITROGEN AND SULFUR – 9701/22/O/N/20/Q3(a)** Syllabus 12.1
....
(a) Describe the environmental consequences of sulfur dioxide emissions. [2]
- 62 NITROGEN AND SULFUR – 9701/21/O/N/23/Q6(b)** Syllabus 12.1
....
(a) Write the equations for the formation of acid rain from nitrogen dioxide. [1]
- 63 NITROGEN AND SULFUR – 9701/22/F/M/20/Q6(c)** Syllabus 12.1
....
(a) Explain how a catalytic converter removes NO from exhaust gases. [2]
- 64 NITROGEN AND SULFUR – 9701/22/F/M/25/Q2(a)** Syllabus 12.1
....
(a) State a natural source of sulfur dioxide in the atmosphere. [1]
- 65 NITROGEN AND SULFUR – 9701/22/M/J/22/Q6(a)** Syllabus 12.1
" [DIAGRAM: Formation of acid rain from atmospheric pollutants] "
(a) Explain how nitrogen oxides are formed in a car engine. [2]

Question Pack

TOPIC 13 — TOPIC 13.1: AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY

66 ORGANIC CHEMISTRY — 9701/21/M/J/23/Q8(c) Syllabus 13.3

...

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

67 ORGANIC CHEMISTRY — 9701/21/F/M/21/Q3(b) Syllabus 13.1

...

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

68 ORGANIC CHEMISTRY — 9701/23/O/N/24/Q4(c) Syllabus 13.4

...

- (a) Identify the functional groups in an amino acid. [2]

69 ORGANIC CHEMISTRY — 9701/23/M/J/21/Q2(a) Syllabus 13.4

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Describe the oxidation of a primary alcohol. [3]

70 ORGANIC CHEMISTRY — 9701/22/M/J/24/Q6(b) Syllabus 13.1

...

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

71 ORGANIC CHEMISTRY — 9701/22/M/J/19/Q7(a) Syllabus 13.4

...

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

72 ORGANIC CHEMISTRY — 9701/23/M/J/24/Q5(a) Syllabus 13.3

...

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

73 ORGANIC CHEMISTRY — 9701/22/O/N/25/Q5(a) Syllabus 13.1

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Identify the functional groups in an amino acid. [2]

74 ORGANIC CHEMISTRY — 9701/21/M/J/24/Q10(c) Syllabus 13.1

...

- (a) Describe the oxidation of a primary alcohol. [3]

75 ORGANIC CHEMISTRY — 9701/21/O/N/22/Q2(b) Syllabus 13.4

...

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

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- 76 ORGANIC CHEMISTRY – 9701/23/F/M/20/Q10(c)** Syllabus 13.1
...
(a) Explain the difference between structural isomerism and stereoisomerism. [2]
- 77 ORGANIC CHEMISTRY – 9701/21/F/M/20/Q1(a)** Syllabus 13.3
"Fig. 3.2 shows the skeletal formula of an organic molecule."
(a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]
- 78 ORGANIC CHEMISTRY – 9701/21/O/N/23/Q7(c)** Syllabus 13.4
...
(a) Identify the functional groups in an amino acid. [2]
- 79 ORGANIC CHEMISTRY – 9701/22/F/M/22/Q1(b)** Syllabus 13.1
...
(a) Describe the oxidation of a primary alcohol. [3]
- 80 ORGANIC CHEMISTRY – 9701/23/F/M/22/Q2(c)** Syllabus 13.1
...
(a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]
- 81 ORGANIC CHEMISTRY – 9701/21/M/J/19/Q3(b)** Syllabus 13.2
"Fig. 3.2 shows the skeletal formula of an organic molecule."
(a) Explain the difference between structural isomerism and stereoisomerism. [2]
- 82 ORGANIC CHEMISTRY – 9701/23/O/N/25/Q5(b)** Syllabus 13.4
...
(a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]
- 83 ORGANIC CHEMISTRY – 9701/22/M/J/21/Q6(b)** Syllabus 13.2
...
(a) Identify the functional groups in an amino acid. [2]
- 84 ORGANIC CHEMISTRY – 9701/21/F/M/20/Q4(c)** Syllabus 13.1
...
(a) Describe the oxidation of a primary alcohol. [3]
- 85 ORGANIC CHEMISTRY – 9701/24/F/M/22/Q4(c)** Syllabus 13.1
"Fig. 3.2 shows the skeletal formula of an organic molecule."
(a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

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86 ORGANIC CHEMISTRY – 9701/23/F/M/23/Q8(c) Syllabus 13.1

- (a) Explain the difference between structural isomerism and stereoisomerism. [2]

87 ORGANIC CHEMISTRY – 9701/22/M/J/24/Q10(b) Syllabus 13.1

- (a) State the Cahn-Ingold-Prelog priority rules for E/Z isomerism. [2]

88 ORGANIC CHEMISTRY – 9701/23/M/J/22/Q10(c) Syllabus 13.4

- (a) Identify the functional groups in an amino acid. [2]

89 ORGANIC CHEMISTRY – 9701/22/O/N/22/Q5(c) Syllabus 13.2

"Fig. 3.2 shows the skeletal formula of an organic molecule."

- (a) Describe the oxidation of a primary alcohol. [3]

90 ORGANIC CHEMISTRY – 9701/22/F/M/22/Q10(c) Syllabus 13.1

- (a) Give the IUPAC name for $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$. [1]

Mark Scheme

MARK SCHEME – WEEK 2

Q	MARKING POINTS	MARKS
1(a)	Pale green / apple green (1).	1
2(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
3(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
4(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
5(a)	Solubility increases down the group (1).	1
6(a)	Pale green / apple green (1).	1
7(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
8(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
9(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
10(a)	Solubility increases down the group (1).	1
11(a)	Pale green / apple green (1).	1
12(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
13(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
14(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
15(a)	Solubility increases down the group (1).	1
16(a)	Pale green / apple green (1).	1
17(a)	Stability increases down the group (1); cation radius increases so less polarising power (1); less distortion of carbonate ion (1).	3
18(a)	$2\text{Mg}(\text{NO}_3)_2 \rightarrow 2\text{MgO} + 4\text{NO}_2 + \text{O}_2$ (1 for species, 1 for balancing).	2
19(a)	Barium is more reactive than magnesium (1). Barium reacts vigorously with cold water whereas Mg reacts very slowly (1).	2
20(a)	Solubility increases down the group (1).	1
21(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2

Mark Scheme

MARK SCHEME — WEEK 2 (CONTINUED)

Q	MARKING POINTS	MARKS
22(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
23(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
24(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
25(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
26(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
27(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
28(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
29(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
30(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
31(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
32(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
33(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
34(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
35(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
36(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
37(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
38(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
39(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2
40(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
41(a)	AgCl forms a soluble complex ion $[\text{Ag}(\text{NH}_3)_2]^+$ (1); AgI is less soluble so does not dissolve in dilute NH_3 (1).	2
42(a)	$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$ (1 for species, 1 for balancing).	2
43(a)	AgNO_3 gives coloured precipitates (1); ammonia tests solubility of precipitates to differentiate between Cl^- , Br^- , I^- (1).	2
44(a)	Chlorine kills bacteria / microorganisms (1); making water safe to drink (1).	2

Mark Scheme

MARK SCHEME — WEEK 2 (CONTINUED)

Q	MARKING POINTS	MARKS
45(a)	Boiling points increase down the group (1); due to more electrons so stronger van der Waals' forces (1).	2
46(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
47(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
48(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
49(a)	Volcanoes (1).	1
50(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
51(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
52(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
53(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
54(a)	Volcanoes (1).	1
55(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
56(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
57(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
58(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
59(a)	Volcanoes (1).	1
60(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
61(a)	Forms acid rain (1); which lowers pH of lakes / damages buildings (1).	2
62(a)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ (1).	1
63(a)	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$ (1); Pt/Pd/Rh catalyst lowers activation energy (1).	2
64(a)	Volcanoes (1).	1
65(a)	High temperature in engine (1); causes nitrogen and oxygen from the air to react (1).	2
66(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2

Mark Scheme

MARK SCHEME — WEEK 2 (CONTINUED)

Q	MARKING POINTS	MARKS
67(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
68(a)	Amine / NH ₂ (1); Carboxylic acid / COOH (1).	2
69(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
70(a)	butan-2-ol (1).	1
71(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
72(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
73(a)	Amine / NH ₂ (1); Carboxylic acid / COOH (1).	2
74(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
75(a)	butan-2-ol (1).	1
76(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
77(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
78(a)	Amine / NH ₂ (1); Carboxylic acid / COOH (1).	2
79(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
80(a)	butan-2-ol (1).	1
81(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
82(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2
83(a)	Amine / NH ₂ (1); Carboxylic acid / COOH (1).	2
84(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
85(a)	butan-2-ol (1).	1
86(a)	Structural isomers have different connectivity of atoms (1); stereoisomers have same connectivity but different spatial arrangement (1).	2
87(a)	Compare atomic number of atoms attached to double bond carbons (1); higher atomic number has higher priority (1).	2

Mark Scheme

MARK SCHEME — WEEK 2 (CONTINUED)

Q	MARKING POINTS	MARKS
88(a)	Amine / NH_2 (1); Carboxylic acid / COOH (1).	2
89(a)	Oxidised to aldehyde (1); then to carboxylic acid (1); using acidified potassium dichromate(VI) (1).	3
90(a)	butan-2-ol (1).	1